The local convergence rate of affine-invariant Gaussian variational inference: one scalar, and where it is hard

Abstract

This note isolates the quantitative content of the local convergence rate of the affine-invariant Gaussian variational-inference flow under $\mu I \preceq \nabla^2 V \preceq \mu' I$. After convexity settles, once and for all, that the linearized generator contracts, the entire rate is governed by a single scalar—the operator norm $\Lambda = \lambda_{\max}(\widetilde{G} - \mathbf{1}\mathbf{1}^\top)$ of a second-moment matrix—which enters the rate reciprocally. We give a self-contained derivation of this reduction, catalogue the bounds on Λ that are currently available (a dimension-free $O(\log \kappa)$ in the structured cases, a dimension-dependent $O(\log \kappa)$ in general, and a dimension-free but slow $O(\kappa)$ from Bochner), and prove that the dimension dependence in the general bound is an artifact of dropping integrability: for the relaxed class of admissible matrix fields, Λ provably grows with dimension. The sharp dimension-free rate for general potentials is therefore equivalent to a single conjectured operator bound, which we state and delimit.

1 Setup and standing assumptions

Let $\pi \propto e^{-V}$ on $\mathbb{R}^{N_\theta}$ with $V \in C^4$ strongly convex,

$$\mu I \ \preceq\ \nabla^2 V(\theta) \ \preceq\ \mu' I \qquad \text{for all } \theta, \qquad \kappa := \mu'/\mu. \tag{1}$$

Gaussian variational inference fits $\rho_a = \mathcal{N}(m, C)$, $a = (m, C)$, by an affine-invariant (natural-gradient) flow whose unique stationary point $a_\star$ is the best Gaussian fit. After an affine reparametrization we normalize $a_\star = (0, I)$, which is equivalent to the two moment conditions, holding under $z \sim \mathcal{N}(0, I)$,

$$\mathbb{E}[\nabla V(z)] = 0, \qquad \mathbb{E}[W(z)] = I, \qquad W(z) := \nabla^2 V(z). \tag{2}$$

Throughout, expectations are under $z \sim \mathcal{N}(0, I_{N_\theta})$, $\langle A, B\rangle_F = \operatorname{tr}(A^\top B)$ is the Frobenius inner product, and $\operatorname{Sym}(N_\theta)$ is the space of symmetric matrices. By (1) and (2) we may take $\mu \le 1 \le \mu'$. We write $V_{abc} = \partial_a\partial_b\partial_c V$ and $V_{abcd} = \partial_a\partial_b\partial_c\partial_d V$; all such tensors are fully symmetric.

2 The linearized rate as a Rayleigh quotient

Linearizing the flow at $a_\star$ in coordinates (u, X) ($m = u$, $C = I + X$, $X \in \operatorname{Sym}(N_\theta)$) produces a self-adjoint generator whose slowest mode is a Rayleigh quotient. We recall the result and fix notation; the derivation (Bonnet–Price differentiation of the moment maps, then Taylor expansion) is standard. Define the linear maps on $\operatorname{Sym}(N_\theta)$ and $\mathbb{R}^{N_\theta}$

$$\mathcal{T}[X]_k := \mathbb{E}[\operatorname{tr}(XW)\, z_k] = \sum_{ij} \mathbb{E}[V_{ijk}]\, X_{ij}, \qquad \mathcal{H}[X]_{ij} := \mathbb{E}[W_{ij}\,(z^\top X z - \operatorname{tr} X)] = \sum_{kl} \mathbb{E}[V_{ijkl}]\, X_{kl}, \tag{3}$$

where the second equalities use Stein's identity $\mathbb{E}[g\,z_i] = \mathbb{E}[\partial_i g]$ iteratively. The map $\mathcal{H}$ is self-adjoint on $\mathrm{Sym}(N_\theta)$. On diagonal arguments we use the matrices

$$\widetilde{T}_{ki} := \mathbb{E}[V_{kii}], \qquad \widetilde{G}_{ij} := \mathbb{E}[W_{ii}\,z_j^2], \qquad \text{so} \qquad \widetilde{G}_{ij} - 1 = \mathbb{E}[V_{iijj}], \tag{4}$$

the last identity from Stein and $\mathbb{E}[W_{ii}] = 1$.

Proposition 1 (Linearized rate; recalled)**.** *The largest eigenvalue $\lambda_{\star,\max} < 0$ of the linearized generator satisfies*

$$-\lambda_{\star,\max} \;=\; \min_{(u,X)\neq 0} \frac{Q(u,X)}{\alpha\|u\|^2 + \|X\|_F^2}, \qquad Q(u,X) := \|u\|^2 + u^\top \mathcal{T}[X] + \tfrac{1}{2}\|X\|_F^2 + \tfrac{1}{4}\operatorname{tr}(X\mathcal{H}[X]), \tag{5}$$

where $\alpha > 0$ is the flow's mean/covariance time-scale ratio. By orthogonal invariance of (5) it suffices to minimize over diagonal $X = \operatorname{diag}\lambda$.

For diagonal $X = \operatorname{diag}\lambda$, (3)–(4) give $\mathcal{T}[\operatorname{diag}\lambda] = \widetilde{T}\lambda$ and $\operatorname{tr}(\operatorname{diag}\lambda\,\mathcal{H}[\operatorname{diag}\lambda]) = \lambda^\top(\widetilde{G} - \mathbf{1}\mathbf{1}^\top)\lambda$, so completing the square in u,

$$Q(u, \operatorname{diag}\lambda) = \big\|u + \tfrac{1}{2}\widetilde{T}\lambda\big\|^2 + \tfrac{1}{4}\,\lambda^\top\big[\,2I + (\widetilde{G} - \mathbf{1}\mathbf{1}^\top) - \widetilde{T}^\top\widetilde{T}\,\big]\lambda. \tag{6}$$

The two questions. Equation (6) exposes the structure of the problem. The *sign* of the λ-form—whether the flow contracts at all—is the inequality $\widetilde{T}^\top\widetilde{T} \preceq 2I + (\widetilde{G} - \mathbf{1}\mathbf{1}^\top)$. Given the sign, the *rate* is controlled by the operator norm of $\widetilde{G} - \mathbf{1}\mathbf{1}^\top$. Sections 4–5 settle the sign unconditionally and reduce the rate to that operator norm; Section 6 is devoted to bounding it. It is convenient to name the central tensor contraction: for $S, S' \in \mathrm{Sym}(N_\theta)$,

$$B(S,S') := \sum_{abcd} S_{ab}S'_{cd}\,\mathbb{E}[V_{abcd}] = \langle \mathcal{H}[S], S'\rangle_F, \qquad \Lambda := \lambda_{\max}(\widetilde{G} - \mathbf{1}\mathbf{1}^\top) = \sup_{\substack{S=\operatorname{diag}\lambda\\ \|S\|_F=1}} B(S,S), \tag{7}$$

the last equality because $B(\operatorname{diag}\lambda, \operatorname{diag}\lambda) = \lambda^\top(\widetilde{G} - \mathbf{1}\mathbf{1}^\top)\lambda$. The basis-free quantity $\sup_{\|S\|_F=1} B(S,S) = \lambda_{\max}(\mathcal{H})$ equals the supremum of Λ over all orthonormal frames; bounding it is what every estimate below does.

3 The one-dimensional tail-cut bound

The single analytic input from convexity is a Gaussian second-moment bound for densities pinched between μ and μ'.

Lemma 2 (Tail-cut bound)**.** *Let $f : \mathbb{R} \to [\mu, \mu']$ be measurable with $\mathbb{E}[f(Z)] = 1$ for $Z \sim \mathcal{N}(0,1)$. Then*

$$\mathbb{E}[f(Z)\,Z^2] \;\le\; D_\kappa := \big(4 + \tfrac{4}{\sqrt{\pi}}\big)(1 + \log\kappa), \qquad \kappa = \mu'/\mu. \tag{8}$$

Proof. If $\kappa \le \sqrt{e}$ then $\mathbb{E}[fZ^2] \le \mu'\mathbb{E}[Z^2] = \mu' \le \kappa \le \sqrt{e} < D_\kappa$. Assume $\kappa > \sqrt{e}$ and set $t_0 := \sqrt{2\log\kappa} \ge 1$. Split at $|Z| = t_0$.

Bulk. On $\{|Z| < t_0\}$, $Z^2 < t_0^2$ and $f \ge 0$, so $\mathbb{E}[fZ^2\,\mathbf{1}_{|Z|<t_0}] \le t_0^2\,\mathbb{E}[f\,\mathbf{1}_{|Z|<t_0}] \le t_0^2\,\mathbb{E}[f] = t_0^2 = 2\log\kappa$.

Tail. Using $f \le \mu'$ and integration by parts, $\mathbb{E}[Z^2\mathbf{1}_{|Z|\ge t_0}] = 2\,t_0\,\varphi(t_0) + 2\,\bar{\Phi}(t_0)$, where $\varphi, \bar{\Phi}$ are the standard normal density and upper tail. By the Mills bound $\bar{\Phi}(t_0) \le \varphi(t_0)/t_0$ and the identity $\mu'\varphi(t_0) = \mu'\,(2\pi)^{-1/2}e^{-\log\kappa} = \mu\,(2\pi)^{-1/2} \le (2\pi)^{-1/2}$,

$$\mathbb{E}[fZ^2\,\mathbf{1}_{|Z|\ge t_0}] \le \mu'\big(2t_0\varphi(t_0) + 2\varphi(t_0)/t_0\big) = \tfrac{2}{\sqrt{2\pi}}\,\mu\,(t_0 + t_0^{-1}) \le \tfrac{2}{\sqrt{2\pi}}(t_0 + t_0^{-1}) \le \tfrac{2}{\sqrt{\pi}}(1 + \log\kappa),$$

the last step using $t_0^{-1} \le 1$, $\frac{2}{\sqrt{2\pi}} t_0 = \frac{2}{\sqrt{\pi}} \sqrt{\log \kappa}$, and $\sqrt{\log\kappa} + \frac{1}{\sqrt{2}} \le 1 + \log\kappa$ for $\log\kappa \ge \frac{1}{2}$. Adding, $\mathbb{E}[fZ^2] \le 2\log\kappa + \frac{2}{\sqrt{\pi}}(1+\log\kappa) \le (2 + \frac{2}{\sqrt{\pi}})(1+\log\kappa) \le D_\kappa$. □

A matching lower-bound construction shows $\sup_f \mathbb{E}[fZ^2] = \Theta(\log\kappa)$, so the $\log\kappa$ in (8) is the correct order.

4 Bochner identity and the matrix Schur inequality (the sign)

Lemma 3 (Bochner identity)**.** *For $u \in \mathbb{R}^{N_\theta}$, $X \in \mathrm{Sym}(N_\theta)$, and $\eta := u + Xz$,*

$$\mathbb{E}[\eta^\top W(z)\,\eta] = \|u\|^2 + 2\,u^\top \mathcal{T}[X] + \|X\|_F^2 + \mathrm{tr}(X\,\mathcal{H}[X]). \tag{9}$$

In particular, for $X = \mathrm{diag}\,\lambda$ and $\eta = u + \mathrm{diag}(\lambda)z$,

$$\mathbb{E}[\eta^\top W\eta] = \|u\|^2 + 2\,u^\top \widetilde{T}\lambda + \|\lambda\|^2 + \lambda^\top(\widetilde{G} - \mathbf{1}\mathbf{1}^\top)\lambda. \tag{10}$$

Proof. Expand $\eta^\top W\eta = u^\top Wu + 2u^\top W(Xz) + (Xz)^\top W(Xz)$ and take expectations. The first term is $u^\top\mathbb{E}[W]u = \|u\|^2$ by (2). For the cross term, Stein gives $\mathbb{E}[W_{ab}z_c] = \mathbb{E}[V_{abc}]$, so $2u^\top\mathbb{E}[WXz] = 2u^\top\mathcal{T}[X]$. For the quadratic term, Stein applied twice gives $\mathbb{E}[W_{ab}z_cz_d] = \mathbb{E}[V_{abcd}] + \delta_{cd}\delta_{ab}$, whence $\mathbb{E}[(Xz)^\top W(Xz)] = \|X\|_F^2 + \mathrm{tr}(X\mathcal{H}[X])$ by full symmetry of $\nabla^4 V$. Specializing $X = \mathrm{diag}\,\lambda$ via (4) gives (10). □

Theorem 4 (Matrix Schur inequality)**.** *Under (2) and $W \succeq 0$,*

$$\widetilde{T}^\top\widetilde{T} \;\preceq\; I + (\widetilde{G} - \mathbf{1}\mathbf{1}^\top). \tag{11}$$

The bound is dimension-free and uses only convexity of V.

Proof. For $\eta = u + \mathrm{diag}(\lambda)z$ the pointwise inequality $\eta^\top W(z)\eta \ge 0$ holds since $W(z) \succeq 0$. Taking expectations and using (10),

$$0 \le \|u\|^2 + 2u^\top\widetilde{T}\lambda + \|\lambda\|^2 + \lambda^\top(\widetilde{G} - \mathbf{1}\mathbf{1}^\top)\lambda \qquad \text{for all } (u,\lambda),$$

i.e. the block matrix $\begin{pmatrix} I & \widetilde{T} \\ \widetilde{T}^\top & I + (\widetilde{G} - \mathbf{1}\mathbf{1}^\top)\end{pmatrix} \succeq 0$. Its upper-left block is $I \succ 0$; the Schur complement criterion gives $I + (\widetilde{G} - \mathbf{1}\mathbf{1}^\top) - \widetilde{T}^\top\widetilde{T} \succeq 0$. □

Theorem 4 gives $\widetilde{T}^\top\widetilde{T} \preceq 2I + (\widetilde{G} - \mathbf{1}\mathbf{1}^\top)$ with room to spare, so the λ-form in (6) is positive definite: the flow contracts exponentially, unconditionally and in every dimension. What remains is the rate.

5 The rate in terms of Λ

Theorem 5 (Rate reduction)**.** *Suppose $\widetilde{G} - \mathbf{1}\mathbf{1}^\top \preceq \Lambda\, I$ for some $\Lambda \ge \mu - 1$. Then*

$$-\lambda_{\star,\max} \;\ge\; \frac{1}{\alpha(\Lambda+2)+4}. \tag{12}$$

Proof. By Proposition 1 and (6), $-\lambda_{\star,\max}$ is the largest ν for which $M_d - \nu D \succeq 0$, where

$$M_d = \begin{pmatrix} I & \widetilde{T}/2 \\ \widetilde{T}^\top/2 & I/2 + (\widetilde{G} - \mathbf{1}\mathbf{1}^\top)/4 \end{pmatrix}, \qquad D = \begin{pmatrix} \alpha I & 0 \\ 0 & I \end{pmatrix}.$$

For $0 < \nu < 1/\alpha$ the block $(1 - \nu\alpha)I \succ 0$, and by the Schur complement criterion $M_d - \nu D \succeq 0$ iff

$$(\tfrac{1}{2} - \nu)I + \tfrac{1}{4}(\widetilde{G} - \mathbf{1}\mathbf{1}^\top) \;\succeq\; \frac{\widetilde{T}^\top \widetilde{T}}{4(1-\nu\alpha)}.$$

Bounding $\widetilde{T}^\top \widetilde{T} \preceq I + (\widetilde{G} - \mathbf{1}\mathbf{1}^\top)$ by Theorem 4, a sufficient condition is

$$\Big[(\tfrac{1}{2} - \nu) - \tfrac{1}{4(1-\nu\alpha)}\Big] I \;\succeq\; \frac{\nu\alpha}{4(1-\nu\alpha)}\,(\widetilde{G} - \mathbf{1}\mathbf{1}^\top),$$

and since $\widetilde{G} - \mathbf{1}\mathbf{1}^\top \preceq \Lambda I$ with a nonnegative coefficient, it suffices that $(\frac{1}{2} - \nu) - \frac{1}{4(1-\nu\alpha)} \geq \frac{\nu\alpha}{4(1-\nu\alpha)}\Lambda$. Multiplying by $4(1-\nu\alpha) > 0$ and simplifying yields the quadratic

$$4\alpha\,\nu^2 - B\,\nu + 1 \geq 0, \qquad B := \alpha(\Lambda + 2) + 4.$$

Since $\Lambda \geq \mu - 1 \geq -1$ we have $B \geq \alpha + 4$, hence $B^2 \geq (\alpha + 4)^2 \geq 8\alpha$, which gives $\sqrt{B^2 - 16\alpha} \leq B - 8\alpha/B$ and therefore the smaller root $\nu_- = (B - \sqrt{B^2 - 16\alpha})/(8\alpha) \geq 1/B$. The form is nonnegative on $[0, \nu_-]$, so $-\lambda_{\star,\max} \geq 1/B$. □

Equation (12) is the organizing principle: every bound on Λ is a rate. With $\Lambda \leq D_\kappa - 1$ it gives the sharp $-\lambda_{\star,\max} \geq [\alpha(1 + D_\kappa) + 4]^{-1} \geq [(5 + 4/\sqrt{\pi})(4 + \alpha)(1 + \log\kappa)]^{-1} = \Theta(1/\log\kappa)$; with $\Lambda \leq \mu' - 1$ only $\Theta(1/\kappa)$.

6 Bounds on Λ: the ladder

We first record the operator form of Λ and the obstruction to soft methods, then catalogue the available bounds.

6.1 Operator reformulation

Lemma 6 (Reformulations)**.** *For $S \in \mathrm{Sym}(N_\theta)$, with $\eta = Sz$,*

$$B(S,S) = \mathbb{E}\big[\eta^\top (W(z) - I)\,\eta\big] = \mathrm{Cov}\big(\langle S, W(z)\rangle_F,\; z^\top S z\big), \tag{13}$$

and consequently

$$\sup_{\|S\|_F = 1} B(S,S) = \Big(\sup_{\|S\|_F=1} \mathbb{E}\big[(Sz)^\top W(z)\,(Sz)\big]\Big) - 1. \tag{14}$$

Both identities use Stein's lemma, i.e. $W = \nabla^2 V$.

Proof. The first equality is (9) with $u = 0$, $X = S$, minus $\mathbb{E}[(Sz)^\top (Sz)] = \mathrm{tr}(S^2) = \|S\|_F^2$. For the second, Stein gives $\mathbb{E}[V_{abcd}] = \mathbb{E}[W_{ab} z_c z_d] - \delta_{ab}\delta_{cd}$, so $B(S,S) = \mathbb{E}[\langle S, W\rangle_F (z^\top S z)] - (\mathrm{tr}\, S)^2$; since $\mathbb{E}[\langle S, W\rangle_F] = \mathrm{tr}\, S = \mathbb{E}[z^\top S z]$ this is the stated covariance. Equation (14) follows from the first equality and $\mathbb{E}\|Sz\|^2 = \|S\|_F^2 = 1$. □

Remark 7 (Soft inequalities lose $\sqrt{\kappa}/\log\kappa$). The covariance form and Cauchy–Schwarz, with $\mathrm{Var}(z^\top Sz) = 2\|S\|_F^2$, give $\sup_{\|S\|_F=1} B(S,S) \le \sqrt{2}\,\sup_{\|S\|_F=1}\mathrm{Var}(\langle S,W\rangle_F)^{1/2}$. Even when the variance is controlled it is of order $\mu'-1$: for separable V and $S=\operatorname{diag}\lambda$ one has $\mathrm{Var}(\langle S,W\rangle_F) = \mathrm{Var}(V_0''(Z))$, which can reach $\mu'-1$, while the true value is $\Theta(\log\kappa)$. The gap is the factor $\sqrt{\kappa}/\log\kappa$: Cauchy–Schwarz (like Bochner and Poincaré) discards the Gaussian-tail confinement that produces the $\log\kappa$ of Lemma 2. Any proof of the $\log\kappa$ scaling must retain that truncation structure inside the matrix argument.

6.2 Dimension-free, slow: the Bochner bound

Proposition 8 (Bochner bound)**.** *$\Lambda \le \mu'-1$, dimension-free.*

Proof. By (14) and $W \preceq \mu' I$, $\sup_{\|S\|_F=1} B(S,S) \le \mu'\sup_{\|S\|_F=1}\mathbb{E}\|Sz\|^2 - 1 = \mu'-1$, and $\Lambda \le \sup_{\|S\|_F=1} B(S,S)$. □

6.3 Dimension-free, fast: structured cases (recalled)

The following are established by reduction to Lemma 2. For *separable* $V=\sum_i V_i(\theta_i)$, $\widetilde{G} - \mathbf{1}\mathbf{1}^\top$ is diagonal with entries $\mathbb{E}[V_i''(z_i)z_i^2]-1 \le D_\kappa - 1$; the same holds in the *rotated-separable* and *aligned* cases, where the maximizing frame coincides with the structure of V. In all three, $\Lambda \le D_\kappa - 1$, giving the sharp dimension-free rate $\Theta(1/\log\kappa)$ via Theorem 5.

6.4 Dimension-free, fast: additive-index models

A natural class strictly larger than rotated-separable admits a dimension-free bound governed by a frame quantity.

Proposition 9 (Additive-index reduction)**.** *Let $V(\theta) = \frac{1}{2}\|\theta\|^2 + \sum_{a=1}^r g_a(\hat b_a^\top\theta)$ with unit vectors $\hat b_a$, and profiles satisfying, with $\phi_a := g_a''$ and $\hat\xi_a := \hat b_a^\top z$,*

$$\mathbb{E}[\phi_a(\hat\xi_a)] = 0, \qquad \mu \le 1+\phi_a(\,\cdot\,) \le \mu'.$$

Then for all $S\in\mathrm{Sym}(N_\theta)$,

$$B(S,S) = \sum_{a=1}^r (\hat b_a^\top S\hat b_a)^2\,\mathbb{E}[\phi_a(\hat\xi_a)\,\hat\xi_a^2], \tag{15}$$

and hence, with $A := \big\|\sum_{a=1}^r \hat b_a\hat b_a^\top\big\|_{\mathrm{op}}$ the upper frame bound,

$$\sup_{\|S\|_F=1} B(S,S) \;\le\; (D_\kappa - 1)\,A. \tag{16}$$

Here $1\le A$, with $A=1$ iff the $\hat b_a$ are orthonormal (the rotated-separable case), and $A = r/N_\theta$ for a tight frame. The rate is $\Theta(1/\log\kappa)$ with the constant scaled by A; dimension-free whenever $A=O(1)$, in particular for overcomplete tight frames of bounded redundancy.

Proof. Here $W - I = \sum_a \phi_a(\hat\xi_a)\,\hat b_a\hat b_a^\top$, so by (13), $B(S,S) = \sum_a \mathbb{E}[\phi_a(\hat\xi_a)\,(\hat b_a^\top Sz)^2]$. Fix a, put $c_a := \hat b_a^\top S\hat b_a$, $s_a := S\hat b_a$, so $\hat b_a^\top Sz = s_a^\top z$. Decompose $s_a^\top z = c_a\hat\xi_a + \rho_a$ with $\rho_a := s_a^\top z - c_a\hat\xi_a$; since $\mathrm{Cov}(s_a^\top z, \hat\xi_a) = s_a^\top\hat b_a = c_a$ and $\mathrm{Var}(\hat\xi_a)=1$, ρ_a and $\hat\xi_a$ are jointly Gaussian with zero covariance, hence independent, and $\mathbb{E}[\rho_a]=0$. Expanding the square, the cross term has expectation $2c_a\mathbb{E}[\phi_a(\hat\xi_a)\hat\xi_a]\mathbb{E}[\rho_a]=0$ and the last term $\mathbb{E}[\phi_a(\hat\xi_a)]\mathbb{E}[\rho_a^2]=0$, proving (15).

Let $h_a := 1 + \phi_a \in [\mu, \mu']$ with $\mathbb{E}[h_a(\hat{\xi}_a)] = 1$; by Lemma 2, $\mathbb{E}[\phi_a(\hat{\xi}_a)\hat{\xi}_a^2] = \mathbb{E}[h_a\hat{\xi}_a^2] - 1 \le D_\kappa - 1$. With each $c_a^2 \ge 0$, (15) gives $B(S,S) \le (D_\kappa - 1)\sum_a(\hat{b}_a^\top S\hat{b}_a)^2$. By Cauchy–Schwarz $(\hat{b}_a^\top S\hat{b}_a)^2 \le \|S\hat{b}_a\|^2$, so $\sum_a(\hat{b}_a^\top S\hat{b}_a)^2 \le \operatorname{tr}(S^2\sum_a \hat{b}_a\hat{b}_a^\top) \le A\|S\|_F^2$, which is (16). The lower bound $A \ge 1$ and the orthonormal equality follow from $\sum_a(\hat{b}_1^\top\hat{b}_a)^4 \ge 1$; tightness for tight frames from $S = N_\theta^{-1/2}I$. □

Remark 10. The hypothesis $\mu \le 1 + \phi_a \le \mu'$ is per profile; it coincides with the global ellipticity (1) only at $A = 1$. For $A > 1$ one has $W - I \preceq (\mu' - 1)\sum_a \hat{b}_a\hat{b}_a^\top$, whose top eigenvalue may reach $1 + (\mu' - 1)A > \mu'$, so for overcomplete frames the proposition governs the class specified by per-coordinate curvature bounds.

6.5 Integrability is necessary: failure of the relaxation

The dimension dependence in the general bound below is not an accident of a crude estimate; it is forced once integrability is dropped. Let

$$\mathcal{W} := \left\{\, W : \mathbb{R}^{N_\theta} \to \mathrm{Sym}(N_\theta) \;\middle|\; \mu I \preceq W(z) \preceq \mu' I \text{ a.s.},\;\; \mathbb{E}[W] = I \,\right\}$$

be the convex relaxation obtained by discarding $W = \nabla^2 V$, and for $W \in \mathcal{W}$ let $b_W(S,S) := \mathbb{E}[\langle S, W\rangle_F(z^\top Sz)] - (\operatorname{tr} S)^2$ be the natural extension of $B(S,S)$ (the two coincide on the Hessian cone by Lemma 6, and differ off it).

Proposition 11 (The relaxed supremum is dimension-dependent)**.** *There are absolute constants $c_0 > 0$, $N_0 \in \mathbb{N}$ such that for all $\mu \le 1 \le \mu'$ and all $N_0 \le N_\theta \le \frac{1}{4}\frac{\mu'-\mu}{1-\mu}$,*

$$\sup_{W\in\mathcal{W}}\ \sup_{\|S\|_F=1}\ b_W(S,S) \;\ge\; c_0\,(1-\mu)\,\sqrt{N_\theta}. \tag{17}$$

For $N_\theta \asymp (\mu'-\mu)/(1-\mu)$ this is of order $\sqrt{\mu'-\mu}$, i.e. $\Theta(\kappa^{1/4})$ in the symmetric normalization $\mu = \kappa^{-1/2}$, $\mu' = \kappa^{1/2}$. Since $D_\kappa - 1 = \Theta(\log\kappa)$, the bound $\Lambda \le D_\kappa - 1$ fails on $\mathcal{W}$ throughout $(\log\kappa)^2 \ll N_\theta \lesssim \sqrt{\kappa}$.

Proof. Write $\hat{z} := z/\|z\|$, $P(z) := \hat{z}\hat{z}^\top$, and for a radius t set $W(z) := \mu I + (\mu'-\mu)\,\mathbf{1}_{\{\|z\|^2\ge t\}}\,P(z)$. Then $\mu I \preceq W \preceq \mu' I$. By rotational invariance, $\mathbb{E}[\mathbf{1}_{\{\|z\|^2\ge t\}}\hat{z}\hat{z}^\top] = \frac{p}{N_\theta}I$ with $p := \Pr(\|z\|^2 \ge t)$, so $\mathbb{E}[W] = I$ iff $p = (1-\mu)N_\theta/(\mu'-\mu)$; the hypothesis gives $p \le \frac{1}{4}$, hence t exceeds the median of $\chi^2_{N_\theta}$ and $t \ge N_\theta$.

Take $S := N_\theta^{-1/2}I$, $\|S\|_F = 1$. With $\chi := \|z\|^2$, $\langle S, W\rangle_F = N_\theta^{-1/2}(\mu N_\theta + (\mu'-\mu)\mathbf{1}_{\{\chi\ge t\}})$ and $z^\top Sz = N_\theta^{-1/2}\chi$, so

$$b_W(S,S) = \mu N_\theta + \tfrac{\mu'-\mu}{N_\theta}\mathbb{E}[\chi\,\mathbf{1}_{\{\chi\ge t\}}] - N_\theta = (1-\mu)(t-N_\theta) + \tfrac{\mu'-\mu}{N_\theta}\mathbb{E}[(\chi-t)_+] \ge (1-\mu)(t-N_\theta),$$

using $\mathbb{E}[\chi\mathbf{1}_{\{\chi\ge t\}}] = tp + \mathbb{E}[(\chi-t)_+]$ and $\frac{\mu'-\mu}{N_\theta}tp = (1-\mu)t$. Finally, by Berry–Esseen for $\chi^2_{N_\theta} = \sum_i z_i^2$ there is an absolute $c_0 > 0$ and N_0 with $t \ge N_\theta + c_0\sqrt{N_\theta}$ whenever $p \le \frac{1}{4}$ and $N_\theta \ge N_0$, giving (17). □

The extremizer is admissible but not a Hessian: the two representations in Lemma 6 disagree on it. Hence removing the dimension dependence requires the Hessian structure—the compatibility the contractions $\mathbb{E}[V_{iijj}]$ inherit from a single potential—and cannot follow from (1)–(2) alone.

6.6 Dimension-dependent, fast: general V (recalled)

Bounding the off-diagonal of $\widetilde{G} - \mathbf{1}\mathbf{1}^\top$ entrywise (via Lemma 2 and its polarization) and summing with the ℓ^1/ℓ^2 inequality $\|\lambda\|_1 \le \sqrt{N_\theta}\,\|\lambda\|_2$ gives, for any V and all N_θ,

$$\Lambda \;\le\; \tfrac{2N_\theta+1}{3}(D_\kappa - 1) + \tfrac{N_\theta-1}{3}(1-\mu) \;=\; O_{N_\theta}(\log\kappa), \tag{18}$$

hence $-\lambda_{\star,\max} \ge \Omega_{N_\theta}(1/\log\kappa)$. The dimension factor is exactly the ℓ^1/ℓ^2 loss flagged by Proposition 11.

6.7 The conjectured dimension-free bound

Conjecture 12 (Operator-norm bound). $\Lambda = \lambda_{\max}(\widetilde{G} - \mathbf{1}\mathbf{1}^\top) \le D_\kappa - 1$, *dimension-free; equivalently* $\sup_{\|S\|_F=1} B(S,S) \le D_\kappa - 1$.

By Theorem 5, Conjecture 12 yields the sharp dimension-free rate $\Theta(1/\log\kappa)$ for every V.

7 Summary: the rate ladder and contraction

Through (12), the status of the local rate is the following hierarchy, in increasing strength of the required structural input.

1. **Separable, rotated-separable, aligned V (unconditional).** $\Lambda \le D_\kappa - 1$ (Section 6.3), rate $\Theta(1/\log\kappa)$.
2. **Additive-index V, bounded frame bound (unconditional).** $\Lambda \le (D_\kappa - 1)A$ (Proposition 9), rate $\Theta(1/\log\kappa)$ with constant A; dimension-free when $A = O(1)$.
3. **General V, fixed dimension (unconditional).** $\Lambda = O_{N_\theta}(\log\kappa)$ (18), rate $\Omega_{N_\theta}(1/\log\kappa)$.
4. **General V, dimension-free (unconditional).** $\Lambda \le \mu' - 1$ (Proposition 8), rate $\Theta(1/\kappa)$ only.
5. **General V (conditional).** $\Lambda \le D_\kappa - 1$ (Conjecture 12), rate $\Theta(1/\log\kappa)$, dimension-free.

The fast rate is available in any fixed dimension (item 3) and dimension-free in every structured case (items 1–2); the single open box is general V, dimension-free, fast—the gap between the κ of item 4 and the $\log\kappa$ of item 5.

From the linear rate to contraction. Whichever bound on Λ holds, the resulting linear rate $\gamma_{\mathrm{loc}} = [\alpha(\Lambda+2)+4]^{-1}$ transfers to genuine nonlinear contraction in a neighborhood of $a_\star$ by a standard Lyapunov/Taylor argument: there is $\delta_0 > 0$ with $\|a_t - a_\star\|^2 \le e^{-\gamma_{\mathrm{loc}} t}\,\|a_0 - a_\star\|^2$ for $\|a_0 - a_\star\| \le \delta_0$. The contraction rate inherits the Λ-regime above; the neighborhood radius δ_0 carries a separate, milder dependence on N_θ through the Frobenius norms of $\mathbb{E}[\nabla^3 V]$ and $\mathbb{E}[\nabla^4 V]$, present already in the separable case and unrelated to the operator-norm question.

Open problem. For a general strongly convex V, removing the N_θ-dependence from the fast rate is equivalent to Conjecture 12: a dimension-free $O(\log\kappa)$ bound on $\Lambda = \sup_{\|S\|_F=1} \mathbb{E}[(Sz)^\top W(Sz)] - 1$ that uses the Hessian structure essentially. Proposition 11 shows ellipticity and normalization alone do not suffice; Proposition 9 settles the additive-index class through the frame bound. A natural weaker sub-target, strictly implied by but plausibly easier than Conjecture 12, is the Hessian variance bound $\sup_{\|S\|_F=1} \mathrm{Var}(\langle S, W\rangle_F) \le \mu' - 1$, which by Remark 7 would already give the dimension-free rate $\Theta(1/\sqrt{\kappa})$—between items 4 and 5—and which fails for the relaxation $\mathcal{W}$, consistent with integrability being the operative constraint.